BIO101 MCQS 4

63

1. **What is the primary purpose of vaccination?**
    - A. To provide immediate treatment for infections
    - B. To stimulate the immune system to produce specific antibodies against pathogens
    - C. To cure diseases
    - D. To replace damaged cells

    Answer: B. To stimulate the immune system to produce specific antibodies against pathogens

2. **What type of immunity is stimulated by vaccination to protect against pathogens?**
    - A. Innate immunity
    - B. Humoral immunity
    - C. Passive immunity
    - D. Autoimmune immunity

    Answer: B. Humoral immunity

3. **What are antibodies that neutralize toxins from pathogens called?**
    - A. Antigens
    - B. Antitoxins
    - C. Complement proteins
    - D. Cytokines

    Answer: B. Antitoxins

4. **How do antibodies contribute to the neutralization of pathogens?**
    - A. By directly killing the pathogen
    - B. By neutralizing specific antigens and binding complement to induce pathogen lysis
    - C. By enhancing pathogen replication
    - D. By directly repairing damaged tissues

    Answer: B. By neutralizing specific antigens and binding complement to induce pathogen lysis

5. **What is the role of cell-mediated immunity in vaccination?**
    - A. To neutralize toxins
    - B. To produce antibodies
    - C. To provide protection against intracellular pathogens
    - D. To enhance pathogen replication

Answer: C. To provide protection against intracellular pathogens

6. **What is the mechanism by which antibodies lead to the destruction of pathogens?**
    - A. By masking the toxins and preventing them from binding to target cells
    - B. By directly attacking the pathogen's DNA
    - C. By enhancing pathogen entry into cells
    - D. By blocking the pathogen's replication in tissues

    Answer: A. By masking the toxins and preventing them from binding to target cells

7. **Why are intracellular pathogens not neutralized by antibodies alone?**
    - A. Because antibodies cannot enter cells
    - B. Because intracellular pathogens are too large
    - C. Because antibodies cannot bind to intracellular pathogens
    - D. Because antibodies do not produce antitoxins

    Answer: A. Because antibodies cannot enter cells

8. **Which immune response is activated to deal with intracellular pathogens during vaccination?**
    - A. Humoral immunity
    - B. Innate immunity
    - C. Cell-mediated immunity
    - D. Passive immunity

    Answer: C. Cell-mediated immunity

9. **What happens to the toxins produced by pathogens when neutralized by antitoxins?**
    - A. They become more active
    - B. They are masked and cannot bind to their specific receptors on target cells
    - C. They are replicated by the immune system
    - D. They enhance the immune response

    Answer: B. They are masked and cannot bind to their specific receptors on target cells

10. **What type of immunity is often involved in the protection provided by vaccines against extracellular pathogens?**
    - A. Cell-mediated immunity
    - B. Passive immunity
    - C. Humoral immunity
    - D. Innate immunity

    Answer: C. Humoral immunity

64

1. **What is the primary characteristic of active immunization?**
    - A. It provides immediate immunity without exposure to antigens.
    - B. It involves the induction of immunity after exposure to an antigen.
    - C. It involves the direct injection of pre-made antibodies.
    - D. It relies on the transfer of maternal antibodies to the fetus.

    Answer: B. It involves the induction of immunity after exposure to an antigen.

2. **Which type of immunization involves the generation of antibodies in response to natural infection or vaccination?**
    - A. Passive immunization
    - B. Active immunization
    - C. Artificial immunization
    - D. Natural passive immunization

    Answer: B. Active immunization

3. **How does passive immunization differ from active immunization?**
    - A. Passive immunization requires antigen exposure, while active does not.
    - B. Passive immunization involves the transfer of pre-made antibodies, while active immunization involves the host's own immune response.
    - C. Active immunization provides immediate protection, while passive takes longer.
    - D. Passive immunization is always achieved through vaccination.

    Answer: B. Passive immunization involves the transfer of pre-made antibodies, while active immunization involves the host's own immune response.

4. **What is an example of natural active immunization?**
    - A. Receiving a vaccine for influenza
    - B. Transfer of maternal antibodies through the placenta
    - C. Getting an infection from a microorganism such as the measles virus
    - D. Injection of tetanus antibodies

    Answer: C. Getting an infection from a microorganism such as the measles virus

5. **Which of the following describes artificial active immunization?**
    - A. Exposure to a less virulent or heat-killed form of a microbe to stimulate immune response
    - B. Transfer of antibodies from mother to fetus through the placenta
    - C. Injection of gamma globulins from another individual
    - D. Transfer of antibodies via breast milk to a newborn

Answer: A. Exposure to a less virulent or heat-killed form of a microbe to stimulate immune response

6. **What is an example of natural passive immunization?**
    - A. Injection of immune globulins from another individual
    - B. Administration of vaccines containing attenuated microbes
    - C. Transfer of IgG antibodies from mother to fetus through the placenta
    - D. Receiving antibodies against tetanus from a serum

    Answer: C. Transfer of IgG antibodies from mother to fetus through the placenta

7. **What type of immunization involves the administration of antibodies from other individuals or animals?**
    - A. Natural active immunization
    - B. Artificial passive immunization
    - C. Artificial active immunization
    - D. Natural passive immunization

    Answer: B. Artificial passive immunization

8. **Which form of immunization is used to provide immediate protection against a disease but is short-lived?**
    - A. Natural active immunization
    - B. Artificial active immunization
    - C. Natural passive immunization
    - D. Artificial passive immunization

    Answer: D. Artificial passive immunization

9. **Which type of immunization provides long-lasting immunity through the production of antibodies by the host?**
    - A. Natural passive immunization
    - B. Artificial passive immunization
    - C. Natural active immunization
    - D. Artificial active immunization

    Answer: C. Natural active immunization

10. **What is the role of gamma globulins in passive immunization?**
    - A. To stimulate the host's immune system to produce antibodies
    - B. To provide pre-made antibodies for immediate protection
    - C. To increase the virulence of pathogens
    - D. To enhance the replication of pathogens

    Answer: B. To provide pre-made antibodies for immediate protection

65

1. **What is the main advantage of DNA vaccines?**
    - A. They only provoke humoral immunity.
    - B. They can generate both humoral and cell-mediated responses.
    - C. They have high efficacy in all stages of clinical use.
    - D. They do not require host cell transfection.

    Answer: B. They can generate both humoral and cell-mediated responses.

2. **Which of the following is a characteristic of immunodominant peptides in vaccine development?**
    - A. They are complex to prepare and require multiple steps.
    - B. They are incorporated with MHC to induce both humoral and cell-mediated responses.
    - C. They are used only for inducing cell-mediated immunity.
    - D. They have a low potential for inducing long-lasting immunity.

    Answer: B. They are incorporated with MHC to induce both humoral and cell-mediated responses.

3. **What is a primary challenge with Anti-HIV-DNA vaccines as mentioned in the experimental stage?**
    - A. They are highly effective in preventing HIV infection.
    - B. They have very low efficacy so far.
    - C. They cause high toxicity in clinical trials.
    - D. They are difficult to clone and produce.

    Answer: B. They have very low efficacy so far.

4. **How do anti-idiotype antibodies contribute to vaccine development?**
    - A. They induce both humoral and cell-mediated responses.
    - B. They act against polysaccharide antibodies and provide long-lasting immune responses with memory.
    - C. They are used only for short-term immune responses.
    - D. They enhance the efficacy of DNA vaccines.

    Answer: B. They act against polysaccharide antibodies and provide long-lasting immune responses with memory.

5. **Which type of novel vaccine involves transfection of cloned viral peptides into host cells?**
    - A. Immunodominant peptides
    - B. DNA vaccines
    - C. Anti-idiotype antibodies

- D. Traditional protein-based vaccines

Answer: B. DNA vaccines

6. **What is a common feature of novel vaccines designed to reduce toxicity?**
   - A. They are primarily used in clinical practice with high success rates.
   - B. They are mainly experimental and used for research purposes.
   - C. They exclusively provoke humoral immunity.
   - D. They require multiple booster doses to be effective.

   Answer: B. They are mainly experimental and used for research purposes.

7. **Which of the following vaccine types is known for its simplicity and ease of preparation?**
   - A. DNA vaccines
   - B. Immunodominant peptides
   - C. Anti-idiotype antibodies
   - D. Live attenuated vaccines

   Answer: B. Immunodominant peptides

8. **What is a potential benefit of using anti-idiotype antibodies in vaccines?**
   - A. They are highly specific and do not provide long-term immunity.
   - B. They can induce a memory response, providing long-lasting immunity.
   - C. They are only effective in treating existing infections.
   - D. They have a high risk of toxicity.

   Answer: B. They can induce a memory response, providing long-lasting immunity.

9. **Which novel vaccine type is mentioned as having potential for future clinical use but is currently in experimental stages?**
   - A. DNA vaccines
   - B. Immunodominant peptides
   - C. Anti-idiotype antibodies
   - D. All of the above

   Answer: D. All of the above

10. **In the context of DNA vaccines, what is meant by "transfection into host cells"?**
    - A. Introducing pre-made antibodies into the host.
    - B. Injecting a live attenuated virus to stimulate immunity.
    - C. Inserting cloned viral peptides into the host's cells to generate an immune response.
    - D. Using polysaccharide vaccines to induce a quick immune response.

Answer: C. Inserting cloned viral peptides into the host's cells to generate an immune response.

66

MCQs on Viruses

1. **What does the term "virus" originate from?**
   - A) Greek for "disease"
   - B) Latin for "slimy liquid" or "poison"
   - C) Greek for "pathogen"
   - D) Latin for "contagion"

   Answer: B) Latin for "slimy liquid" or "poison"

2. **What is a key characteristic of viruses?**
   - A) They can replicate independently of host cells.
   - B) They have a complex metabolism similar to bacteria.
   - C) They are obligate intracellular parasites.
   - D) They have a cellular structure like that of eukaryotes.

   Answer: C) They are obligate intracellular parasites.

3. **Which of the following is NOT a component of a virus?**
   - A) Nucleic acid
   - B) Protein coat (capsid)
   - C) Cellular membrane
   - D) Envelope (in some viruses)

   Answer: C) Cellular membrane

4. **What size range do most viruses fall into?**
   - A) 1-5 micrometers
   - B) 20-300 nanometers
   - C) 0.1-2 micrometers
   - D) 500-1000 nanometers

   Answer: B) 20-300 nanometers

5. **Which scientist first described viruses as "poison"?**
   - A) Louis Pasteur
   - B) Robert Koch
   - C) Martinus Beijerinck
   - D) Friedrich Loeffler

Answer: C) Martinus Beijerinck

6. **What is the primary way viruses replicate?**
    - A) Through cell division
    - B) By using their own metabolic machinery
    - C) By invading host cells and using their machinery
    - D) By asexual reproduction

    Answer: C) By invading host cells and using their machinery

7. **Which of the following statements is true about viruses?**
    - A) They have their own metabolism and can reproduce outside host cells.
    - B) They are larger than bacteria and can be seen with optical microscopes.
    - C) They consist of nucleic acid enclosed in a protein coat and sometimes an envelope.
    - D) They contain both DNA and RNA simultaneously.

    Answer: C) They consist of nucleic acid enclosed in a protein coat and sometimes an envelope.

8. **Which type of nucleic acid can viruses have?**
    - A) DNA only
    - B) RNA only
    - C) Either DNA or RNA
    - D) Both DNA and RNA

    Answer: C) Either DNA or RNA

67

MCQs on Unique Properties of Viruses

1. **Which of the following is a unique characteristic of viruses?**
    - A) They can be cultivated easily in laboratory settings.
    - B) They have a complex cellular structure.
    - C) They have both living and non-living characteristics.
    - D) They replicate independently of host cells.

    Answer: C) They have both living and non-living characteristics.

2. **What is a major challenge in studying viruses in the laboratory?**
    - A) They have complex replication machinery.
    - B) They can easily be cultivated in various media.
    - C) They cannot be easily cultivated in laboratory settings.
    - D) They contain their own metabolic systems.

Answer: C) They cannot be easily cultivated in laboratory settings.

3. **Which of the following statements is true about viruses compared to bacteria?**
    - A) Viruses are more complex and have more metabolic functions than bacteria.
    - B) Bacteria replicate using host cell machinery, while viruses replicate independently.
    - C) Bacteria are usually free-living and perform most of their basic metabolic functions themselves.
    - D) Viruses and bacteria are similar in size and complexity.

    Answer: C) Bacteria are usually free-living and perform most of their basic metabolic functions themselves.

4. **Which of the following is NOT a method used for studying viral cultivation in the lab?**
    - A) Tissue culture
    - B) Embryonated egg
    - C) Live animal
    - D) Chemical synthesis

    Answer: D) Chemical synthesis

5. **What is the main component of a virus's structure?**
    - A) A complex cellular membrane
    - B) A large ribosome
    - C) A small genome encased in a protein coat
    - D) A large number of mitochondria

    Answer: C) A small genome encased in a protein coat

6. **What makes prions unique among infectious agents?**
    - A) They are made of RNA.
    - B) They contain a complex genome.
    - C) They are made only of protein and have no genome.
    - D) They are larger than viruses and bacteria.

    Answer: C) They are made only of protein and have no genome.

7. **What is the correct order of the general steps in the viral life cycle?**
    - A) Synthesis of viral proteins, disassembly, replication of the viral genome, reassembly
    - B) Replication of the viral genome, disassembly, synthesis of viral proteins, reassembly
    - C) Disassembly, replication of the viral genome, synthesis of viral proteins, reassembly

- D) Reassembly, replication of the viral genome, disassembly, synthesis of viral proteins

Answer: C) Disassembly, replication of the viral genome, synthesis of viral proteins, reassembly

8. **How do viruses replicate within host cells?**
   - A) By using their own cellular machinery for metabolism.
   - B) By subverting the host cell's transcription and translation machinery.
   - C) By independently replicating their DNA or RNA outside of the host cell.
   - D) By incorporating into the host's mitochondrial DNA.

   Answer: B) By subverting the host cell's transcription and translation machinery

68

MCQs

1. **What is a virus?**
   - A) A microscopic organism that can live independently
   - B) A sub-microscopic particle that can infect living cells
   - C) A type of prokaryotic cell
   - D) A eukaryotic cell

   Answer: B) A sub-microscopic particle that can infect living cells

2. **What is the typical size range of viruses?**
   - A) 0.5–5.0 micrometers (µm)
   - B) 20–300 nanometers (nm)
   - C) 1–2 millimeters
   - D) 5–10 micrometers (µm)

   Answer: B) 20–300 nanometers (nm)

3. **How does the size of a virus compare to that of a prokaryote?**
   - A) Viruses are larger than prokaryotes
   - B) Viruses are similar in size to prokaryotes
   - C) Viruses are much smaller than prokaryotes
   - D) Viruses are much larger than prokaryotes

   Answer: C) Viruses are much smaller than prokaryotes

4. **Which of the following is NOT a characteristic of viruses?**
   - A) They have cell membranes
   - B) They lack cytoplasm and ribosomes

- C) They require a host cell to reproduce
- D) They contain genetic material

Answer: A) They have cell membranes

5. **What is the structure of a simple virus?**
   - A) A protective protein coat and nucleic acid genome
   - B) A lipid bilayer and nucleic acid genome
   - C) A cell membrane and ribosomes
   - D) A cell wall and mitochondria

   Answer: A) A protective protein coat and nucleic acid genome

6. **Which types of nucleic acids can viruses contain?**
   - A) Deoxyribonucleic Acid (DNA) or Ribonucleic Acid (RNA)
   - B) Only Deoxyribonucleic Acid (DNA)
   - C) Only Ribonucleic Acid (RNA)
   - D) Proteins and carbohydrates

   Answer: A) Deoxyribonucleic Acid (DNA) or Ribonucleic Acid (RNA)

7. **Why do viruses need a host cell?**
   - A) To gain energy
   - B) To replicate their nucleic acids and synthesize proteins
   - C) To digest food
   - D) To maintain their metabolism

   Answer: B) To replicate their nucleic acids and synthesize proteins

8. **What additional components do complex viruses have compared to simple viruses?**
   - A) Cell membranes
   - B) Mitochondria and chloroplasts
   - C) Lipids, polysaccharides, and trace elements
   - D) Ribosomes and cytoplasm

   Answer: C) Lipids, polysaccharides, and trace elements

69

MCQs

1. **What is the main structural difference between enveloped and non-enveloped viruses?**
   - A) Enveloped viruses have a nucleocapsid, while non-enveloped viruses do not.

- B) Non-enveloped viruses have an envelope and spikes, while enveloped viruses only have a nucleocapsid.
- C) Enveloped viruses have an envelope and a nucleocapsid, while non-enveloped viruses only have a nucleocapsid.
- D) Enveloped viruses lack nucleic acid, whereas non-enveloped viruses have nucleic acid.

Answer: C) Enveloped viruses have an envelope and a nucleocapsid, while non-enveloped viruses only have a nucleocapsid.

2. **What is the function of spikes on some viruses?**
   - A) To protect the virus from host defenses
   - B) To assist in the attachment to the host cell
   - C) To replicate the viral nucleic acid
   - D) To provide metabolic energy to the virus

   Answer: B) To assist in the attachment to the host cell

3. **What type of symmetry do helical viruses exhibit?**
   - A) Icosahedral symmetry
   - B) Helical symmetry
   - C) Complex symmetry
   - D) Spherical symmetry

   Answer: B) Helical symmetry

4. **Which of the following viruses is an example of a helical virus?**
   - A) Tobacco mosaic virus
   - B) Adenovirus
   - C) Herpesvirus
   - D) Bacteriophage T4

   Answer: A) Tobacco mosaic virus

5. **How do polyhedral viruses differ from helical viruses in terms of structure?**
   - A) Polyhedral viruses have a coiled, ribbon-like structure, while helical viruses have a spherical structure.
   - B) Polyhedral viruses have a repeating protein subunits forming a shell, while helical viruses have a coiled structure with nucleic acid.
   - C) Polyhedral viruses are enveloped, while helical viruses are non-enveloped.
   - D) Polyhedral viruses have spikes, while helical viruses do not.

   Answer: B) Polyhedral viruses have a repeating protein subunits forming a shell, while helical viruses have a coiled structure with nucleic acid.

6. **Which viruses are categorized as having complex shapes?**

- A) Viruses with helical or polyhedral capsids
- B) Viruses with spherical or rod-like shapes
- C) Poxviruses and large bacteriophages
- D) Simple helical viruses

Answer: C) Poxviruses and large bacteriophages

7. **What is a common example of a virus with a polyhedral shape?**
   - A) Tobacco mosaic virus
   - B) Influenza virus
   - C) Rabies virus
   - D) Poxvirus

   Answer: B) Influenza virus

8. **Which virus is known for having a helical symmetry?**
   - A) Human papillomavirus
   - B) Influenza virus
   - C) Adenovirus
   - D) Poxvirus

   Answer: B) Influenza virus

70

Multiple-Choice Questions (MCQs) on Bacteriophages

1. **What is a bacteriophage?**
   - A) A type of bacteria that infects other bacteria
   - B) A virus that infects bacteria
   - C) A type of eukaryotic virus
   - D) A virus that infects plants

   Answer: B) A virus that infects bacteria

2. **Which cycle involves a bacteriophage hijacking a bacterium to produce many phages and then killing the cell?**
   - A) Lysogenic cycle
   - B) Lytic cycle
   - C) Retroviral cycle
   - D) Endocytotic cycle

   Answer: B) Lytic cycle

3. **In the lysogenic cycle, what happens to the bacteriophage DNA?**

- A) It is immediately replicated and released from the host cell.
- B) It integrates into the bacterial chromosome and is copied along with the host DNA.
- C) It destroys the bacterial chromosome and causes cell lysis.
- D) It forms a new virus particle inside the bacterial cell.

Answer: B) It integrates into the bacterial chromosome and is copied along with the host DNA.

4. **Which of the following is NOT a common shape of the bacteriophage capsid?**
   - A) Icosahedral
   - B) Filamentous
   - C) Head-tail
   - D) Spherical

   Answer: D) Spherical

5. **What is the unique feature of the head-tail structure in bacteriophages?**
   - A) It is a shape unique to eukaryotic viruses.
   - B) It consists of a filamentous tail and a spherical head.
   - C) It is a shape unique to bacteriophages and their close relatives.
   - D) It consists of an icosahedral head and a helical tail.

   Answer: C) It is a shape unique to bacteriophages and their close relatives.

6. **Which type of genetic material can a bacteriophage genome contain?**
   - A) Only DNA
   - B) Only RNA
   - C) Either DNA or RNA
   - D) Both DNA and RNA simultaneously

   Answer: C) Either DNA or RNA

7. **Which bacteriophage is used as an example to illustrate the lytic and lysogenic cycles?**
   - A) Lambda (λ) phage
   - B) T4 phage
   - C) MS2 phage
   - D) M13 phage

   Answer: A) Lambda (λ) phage

8. **How many genes can a bacteriophage genome contain?**
   - A) Fewer than four genes or more than several hundred genes
   - B) Exactly 100 genes
   - C) Only 4 genes

- D) Between 10 to 20 genes

Answer: A) Fewer than four genes or more than several hundred genes